

	School: School of Sciences Department: Department of Physics Program: B. Tech. CSE	Year: 1 Semester: 1, 2	Subject Name: Semiconductor and Quantum Physics Name/code: 26SOST101 L-T-P-C: 3-0-0-3 Pre requisite: Intermediate Mathematics and Physics
Course Outcome	Description (At the end of the course, student will be able to)		
C01	Analyze the concept of energy band theory, semiconductor behavior, and superconductivity to differentiate the electrical properties of solid materials		
C02	Apply the knowledge of semiconductor diode to the development of novel optoelectronic devices for engineering applications.		
C03	Explain the wave-particle duality of matter and apply Schrödinger's equation to analyze fundamental quantum phenomena.		
C04	Analyze the principles of laser operation and optical fibre technology to evaluate their applications in modern engineering and optical communication systems.		
Detailed Syllabus (Theory)			
Unit No.	Topics	CO No.	No. of proposed lectures
1	Introduction to Solids: Energy band in solids, Fermi Dirac distribution function and Fermi level, classification: metals, semiconductors, and insulators. Intrinsic and extrinsic semiconductors, concept of holes, carrier concentration. Superconductors: temperature dependence of resistivity and critical field in superconducting materials, Meissner effect.	1	10
2	Semiconductor Devices: Direct and indirect band gaps, p-n junction diode-formation of depletion region, barrier potential, biasing, and I-V characteristics, optoelectronics devices-LEDs, laser diode. Basics of photovoltaics- photovoltaic effect, determination of efficiency of PV cell.	2	10

3	Quantum Physics: Limitations of classical mechanics (Qualitative), de Broglie's hypothesis. Introduction to quantum computing (qualitative), Wave particle duality, Physical significance of the wave function, Time-independent and dependent Schrodinger wave equations, Schrodinger equation for particle in one dimensional box, particle in finite potential, tunneling effect (Qualitative).	3	10
4	Photonics and Optical Communication: Spontaneous and stimulated emission of radiations, Einstein's coefficients, metastable state, population inversion, lasing action, solid state laser (Ruby laser), gas laser (He-Ne laser), semiconductor laser and applications. Principle and construction of an optical fiber, acceptance angle, numerical aperture, types of optical fibers, and applications.	4	10
Textbooks: 1. H.K. Malik and A.K. Singh, 2nd edition 2017, Engineering Physics. 2. S.K. Gupta, Engineering Physics I and II, 2012.			
Reference Books: 1. Arthur Beiser, Concepts of Modern Physics, 7th edition, 2017. 2. S.P. Taneja, Modern Physics for Engineers, 7th edition, 2010, R Chand and Company, India.			

Date: 29/07/2026